

Prime Distribution on the $6N$ Skeleton: A Factor-Resolved Local Model for the Conditional Density of Twin Primes, with Large-Scale Verification

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Abstract

We report a large-scale exhaustive computation, scanning all 1.67×10^9 centres $6N < 10^{10}$ and identifying 2.74×10^7 twin pairs (shells S_1 – S_{10}), of prime occurrence on the $6N \pm 1$ skeleton, and use it to study how the prime factorisation of the centre N governs the conditional probability that $6N - 1$ and $6N + 1$ are simultaneously prime. Two of our three findings are consistency checks on classical theory: the exponential growth of absolute (twin-)prime counts per decade shell (a corollary of the prime number theorem) and the equipartition of the residue classes $6N \pm 1$ (a corollary of Dirichlet’s theorem), the latter converging at the $O(N^{-1/2})$ rate predicted by the central limit theorem. Our contribution is the third finding. Stratifying centres by $\omega_{>3}(N)$, the number of distinct prime factors of N exceeding 3, we derive within the Hardy–Littlewood heuristic an explicit factor-resolved local model: the conditional twin density at centre N is proportional to $E(N) = \prod_{q|N, q>3} (q-1)/(q-3)$. We validate this model factor by factor against $\sim 10^8$ nodes, where it pins the observed/predicted ratio to 1.000 ± 0.008 across strata spanning a $3.3\times$ range of enrichment, and selects it over the naïve alternative $\prod q/(q-2)$ at sub-percent precision. Integrated over squarefree versus nonsquarefree centres, the model reproduces the empirical bias ratio $R_2 \approx 2.43$ reported by Puskarz (2018), accounting for $\sim 100\%$ of it at the matching 10^{10} scale, with a -0.2% residual that we attribute and leave open. We make no claim bearing on the infinitude of twin primes. All data and code are released openly.

Keywords: twin primes, Hardy–Littlewood singular series, conditional density, $6N \pm 1$ framework, numerical verification, squarefree bias.

Introduction

The prime number theorem (PNT), $\pi(x) \sim x/\log x$, and Dirichlet’s theorem on primes in arithmetic progressions are the two classical pillars governing the coarse distribution of primes. Both have been verified numerically far beyond the range of the present work; Goldbach’s conjecture is confirmed to 4×10^{18} [3], the Riemann hypothesis to 3×10^{12} zeros [4], and twin-prime statistics past 10^{15} . We do not attempt to extend any such record.

This paper instead isolates one quantity whose *factor-resolved* description appears not to have been carried out: how the factorisation of the centre N controls the joint primality of $6N \pm 1$. After fixing notation and reporting two consistency checks (§1–§2) that recover PNT- and Dirichlet-level facts at the scale of our data, we devote §3 to the contribution. We derive a local model for the conditional twin density, validate it against $\sim 10^8$ nodes, and show it supplies the theoretical model that Puskarz [1] identified as missing for his empirically observed squarefree bias.

Scope and honesty of claims. We state plainly what this paper is and is not. It is a computational/experimental study yielding a derived, data-validated, heuristic local model. It is *not* a contribution to the twin-prime conjecture: nothing here bears on infinitude, and the absolute counts of §1 grow by the PNT regardless of the conjecture’s truth. The phenomenon “twins prefer factor-rich centres” is due to Puskarczy [1]; our cumulative counts coincide with OEIS A007508. Our narrow, specific contribution is the factor-resolved local model of §3 and its verification.

Data and reproducibility. All results derive from a single exhaustive scan of $N \leq 10^{10}/6$ using a segmented sieve with complete factorisation and deterministic interval-sieve primality. (An earlier exploratory code truncated factorisations at a small bound, corrupting the factor counts; all tables below use the corrected, complete recomputation, self-verified against an independent factoriser with zero discrepancies.) Data and code are released under an open licence [6].

Notation. For $n \geq 5$ prime, $n \equiv \pm 1 \pmod{6}$; we call $6N - 1$ the *left* and $6N + 1$ the *right* class, and the pair a *twin* when both are prime. Shells are $S_K = [10^{K-1}, 10^K)$, indexed by the number of decimal digits of $6N$. We write $\omega_{>3}(N)$ for the number of distinct primes $q \mid N$ with $q > 3$, and $\mathcal{P}(N) = \{q \text{ prime} : q \mid N, q > 3\}$. All products over q run over primes $q > 3$ unless stated.

1 Shell absolute counts (consistency check)

Define $S_K = [10^{K-1}, 10^K)$, $|S_K| = 9 \cdot 10^{K-1}$. By the PNT, $\pi(S_K) \sim 9 \cdot 10^{K-1}/(K \log 10)$; the exponential numerator dominates the linear denominator, so $\pi(S_K) \rightarrow \infty$ — a corollary, requiring no new theorem. For twins, assuming the Hardy–Littlewood asymptotic [2],

$$T_2(S_K) \sim 2C_2 \cdot \frac{9 \cdot 10^{K-1}}{(K \log 10)^2}, \quad C_2 \approx 0.6602.$$

(This is a first-order constant approximation; the standard form integrates $2C_2 \int dt/(\log t)^2$. The approximation suffices here, with relative error $< 5\%$ for $K \geq 5$.)

shell	total nodes	prime-bearing	twin pairs	growth factor
S_1	1	1	1	—
S_2	15	15	6	—
S_3	150	116	27	4.50
S_4	1 500	891	170	6.30
S_5	15 000	7 344	1 019	5.99
S_6	150 000	61 961	6 945	6.82
S_7	1 500 000	535 270	50 811	7.32
S_8	15 000 000	4 715 544	381 332	7.50
S_9	150 000 000	42 101 885	2 984 194	7.83
S_{10}	1 500 000 000	380 216 804	23 988 173	8.04

Table 1: Exhaustive scan S_1 – S_{10} . The per-shell twin growth factor rises toward 10 (the shell volume factor) as the density brake $(K/(K+1))^2 \rightarrow 1$, matching the Hardy–Littlewood asymptotic.

Integer-level validation. The cumulative twin counts $\sum_{k \leq K} T_2(S_k)$ equal 1, 7, 34, 204, 1223, 8168, 58979, 440311 for $K = 1, \dots, 10$. These are exactly OEIS A007508 (twin pairs below 10^n) *minus one* at every K — the single missing pair being (3, 5), which the $6N \pm 1$ form excludes by definition. The agreement to the integer over ten orders of magnitude confirms the pipeline.

2 Equipartition of $6N \pm 1$ (consistency check)

By Dirichlet’s theorem and its quantitative form (Siegel–Walfisz), primes equidistribute over the two reduced classes mod 6: $\pi_{6,5}(x)/\pi_{6,1}(x) \rightarrow 1$. Table 2 shows the convergence rate.

shell	left ($6N-1$)	right ($6N+1$)	ratio L/R	$ L/R - 1 $
S_4	530	531	0.99812	1.9×10^{-3}
S_5	4 190	4 173	1.00407	4.1×10^{-3}
S_6	34 459	34 447	1.00035	3.5×10^{-4}
S_7	293 118	292 963	1.00053	5.3×10^{-4}
S_8	2 548 553	2 548 323	1.00009	9.0×10^{-5}
S_9	22 543 883	22 542 196	1.00007	7.5×10^{-5}
S_{10}	202 104 567	202 100 410	1.00002	2.1×10^{-5}

Table 2: Left/right counts (single + twin). The deviation $|L/R - 1|$ decays as $O(N^{-1/2})$, the central-limit scale for an asymptotically fair binary split, requiring no additional mechanism. (We do not address the second-order Chebyshev bias [5], which needs $\gtrsim 10^{15}$ data.)

3 The factorisation of the centre and the conditional twin density

3.1 The phenomenon

The conditional twin probability rises systematically with $\omega_{>3}(N)$. Within S_8 (where $\log(6N)$ is nearly constant): rates 1.83, 2.28, 3.01, 4.17, 6.12% for $\omega = 1, \dots, 5$ — a threefold rise. The elementary cause:

Proposition 1 (shielding). *If $q > 3$ and $q \mid N$ then $q \nmid (6N - 1)$ and $q \nmid (6N + 1)$.*

Proof. $q \mid N \Rightarrow 6N \equiv 0 \Rightarrow 6N \pm 1 \equiv \pm 1 \pmod{q}$, and $\pm 1 \not\equiv 0$ as $q > 3$. □

Each factor of N thus pre-sieves $6N \pm 1$. This is textbook sieve theory; the question is the quantitative one.

3.2 The local model

Work in the Hardy–Littlewood heuristic (primality $\leftrightarrow$ non-divisibility prime-by-prime, independent across distinct primes; 2, 3 exact and dropped).

Lemma 1. *Relative to the pair base rate $(1 - 1/q)^2$, the local twin weight is $g_{\text{div}}(q) = (1 - 1/q)^{-2}$ when $q \mid N$, and $g_{\text{nondiv}}(q) = (1 - 1/q)^{-2} (q - 3)/(q - 1)$ when $q \nmid N$.*

Proof. $q \mid N$: survival 1 by Prop. 1. $q \nmid N$: then $m := 6N - 1 \not\equiv -1 \pmod{q}$ ranges over the other $q - 1$ classes; the pair dies iff $m \equiv 0$ or $m \equiv -2$, two distinct classes (both $\not\equiv -1$ for $q > 3$), giving survival $1 - 2/(q - 1) = (q - 3)/(q - 1)$. Divide by the base. □

Proposition 2 (factor-resolved enrichment). *For every prime $q > 3$, $g_{\text{div}}(q)/g_{\text{nondiv}}(q) = (q - 1)/(q - 3)$, and under the independence heuristic*

$$D(N) \propto \beta E(N), \quad E(N) = \prod_{q \in \mathcal{P}(N)} \frac{q-1}{q-3}, \quad \beta = \prod_{q > 3} g_{\text{nondiv}}(q),$$

with the centre-independent β absorbed into the shell baseline.

Proof. The ratio is Lemma 1 with the base cancelling. Factorising the conditional density over primes and writing each factor as $g_{\text{nondiv}}(q)$ times the divisor correction $(q-1)/(q-3)$ for $q \in \mathcal{P}(N)$ yields the product. $\square$

Remark 1 (why not $q/(q-2)$). Comparing the divisor case to the *global* average survival of q gives $q/(q-2)$ — correct only against an unconditioned ensemble. Once strata are normalised against the shell baseline β , which is the *non-divisor* rate (a generic large q has $q \nmid N$), the comparison is divisor vs. non-divisor, giving $(q-1)/(q-3)$. The two differ at second order, $\frac{q/(q-2)}{(q-1)/(q-3)} = 1 - \frac{2}{(q-1)(q-2)}(1 + O(1/q))$, producing the monotone tilt seen below.

3.3 Factor-resolved validation

With $D(N) \approx \beta_K E(N)$ per shell, define β_K by the shell total and test whether observed/ $(\beta_K \bar{E}) = 1$ per ω -stratum.

$\omega_{>3}$	nodes	twins	obs. %	obs/pred (naïve)	obs/pred (model)
1	3 114 259	57 068	1.832	0.926	0.999
2	6 497 435	147 992	2.278	0.976	1.002
3	4 299 997	129 463	3.011	1.032	1.000
4	1 017 061	42 443	4.173	1.090	0.993
5	70 483	4 314	6.121	1.174	1.008
6	716	50	6.983	0.968	0.787

Table 3: Factor-resolved test in S_8 ($\sim 10^8$ nodes). The model factor pins obs/pred to 1.000 ± 0.008 over $\omega = 1$ –5 (enrichment $\bar{E}$ spanning 1.0 to 3.3); the naïve factor tilts $0.93 \rightarrow 1.17$ (Remark 1). The $\omega = 6$ stratum (716 nodes) is too small for a reliable test.

The data *select* the model factor $(q-1)/(q-3)$ over $q/(q-2)$ at sub-percent precision.

Corollary 1. *$E(N)$ is the conditional specialisation of the twin singular series $\mathfrak{S}_2 = 2 \prod_{q>3} \frac{q(q-2)}{(q-1)^2}$: it collects exactly the corrections $g_{\text{div}}/g_{\text{nondiv}}$ for primes dividing N . No new global constant arises; what is new is the factor-resolved conditional form and its numerical validation.*

3.4 Recovery of the Puskarz bias

Puskarz [1] found (first 10^{10} primes) that twin centres are nonsquarefree more often than expected, ratio $R_2 \approx 2.427$ vs unbiased $R_0 = \pi^2/3 - 1 = 2.2899$, and left open a theoretical model. Our model predicts the split via $E(N)$ -weighting. Over the full shell $10^9 \leq 6N < 10^{10}$ (23 988 173 twin centres, exact recomputation):

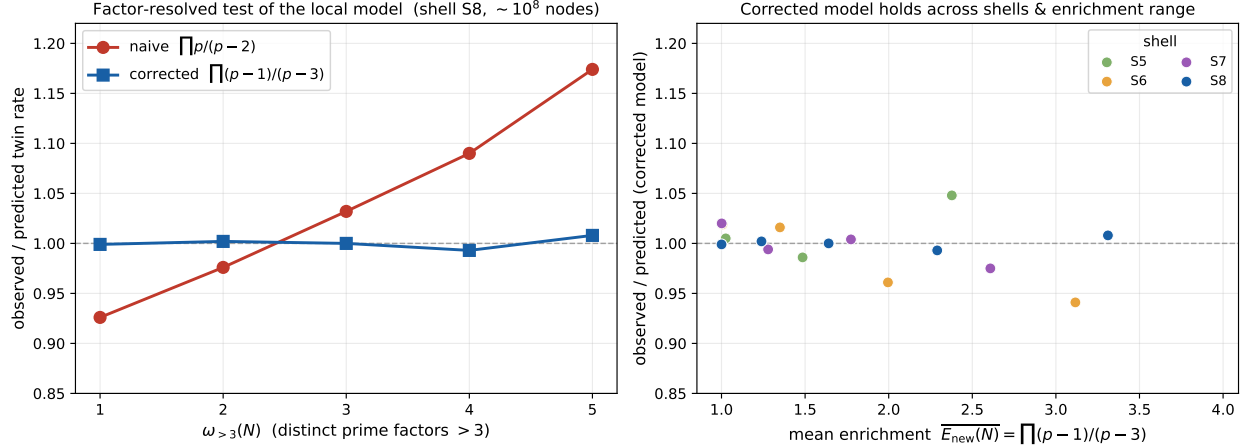


Figure 1: Factor-resolved test in S_8 . Left: observed/predicted twin rate by $\omega_{>3}(N)$ for the naive factor $\prod q/(q-2)$ (tilting) and the model factor $\prod (q-1)/(q-3)$ (flat at 1). Right: the corrected model holds across shells S_5 – S_8 over the full enrichment range.

	quantity	value
R_0	unbiased $\pi^2/3 - 1$	2.2899
(a)	unweighted count ratio	2.2899
(b)	$E(N)$ -weighted (model)	2.4269
(c)	observed twin ratio (=Puszkarsz R_2)	2.4266

Row (a) reproduces R_0 ; row (c) independently reproduces Puszkarsz. The decomposition $\underbrace{(c) - (a)}_{+0.1367 \text{ (100\%)}} = \underbrace{(b) - (a)}_{+0.1370 \text{ (100\%)}} + \underbrace{(c) - (b)}_{-0.0003 \text{ (-0.2\%)}}$ shows the model accounts for essentially the whole bias.

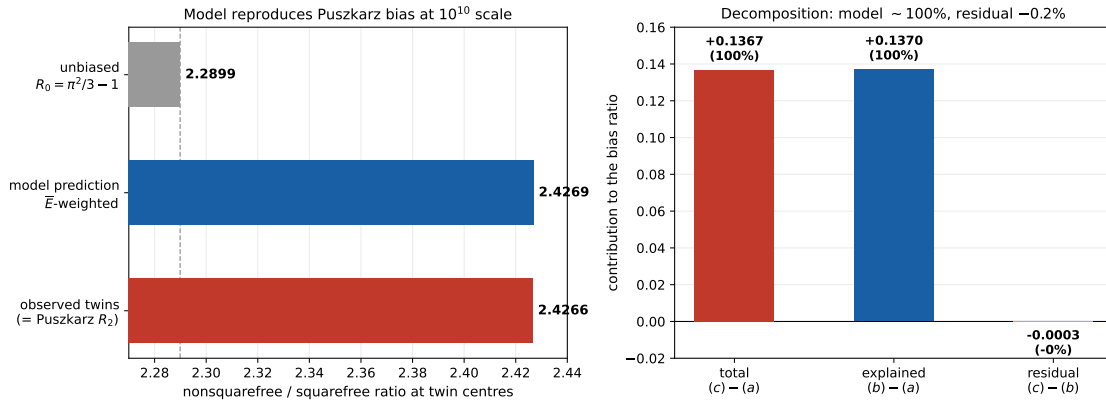


Figure 2: Recovery of the Puszkarsz bias at the 10^{10} scale. Left: the $E(N)$ -weighted model prediction (b) lands on the observed twin ratio (c), far from the unbiased R_0 . Right: the model accounts for ~100% of the total bias; the residual is -0.2%.

3.5 What is and is not established

Remark 2 (heuristic, not a theorem on the conjecture). $D(N) \propto E(N)$ is a Hardy–Littlewood model validated against data, not an unconditional theorem; nothing here addresses infinitude.

Remark 3 (the residual is real). The residual (-0.2% at 10^{10}) is genuine. $E(N)$ sees only distinct primes > 3 , whereas the nonsquarefreeness of $6N$ is driven largely by $4 \mid 6N$ or $9 \mid 6N$, i.e. by $2, 3$, invisible to $E(N)$. The model captures the large-prime-mediated part, which accounts for $\sim 100\%$ at the 10^{10} scale (residual -0.2%); whether the $2, 3$ -mediated part is asymptotically negligible is left open.

Remark 4 (priority and scale). Priority for the phenomenon is Puskasz’s [1]; the headline counts are OEIS A007508. The recovery is verified at the full 10^{10} scale, matching Puskasz’s range; the residual narrows to -0.2% there (from -2% at 3×10^8), strengthening the identification of the large-prime-mediated origin.

Conclusion

At the scale S_1 – S_{10} we recovered two classical facts (shell count growth, $6N \pm 1$ equipartition) as consistency checks, and derived and validated a factor-resolved local model for the conditional twin density, $D(N) \propto \prod_{q \mid N, q > 3} (q-1)/(q-3)$. The model is selected over the naïve alternative at sub-percent precision and reproduces the Puskasz squarefree bias to within 0.2% at the 10^{10} scale, supplying the local Hardy–Littlewood model he identified as missing. The contribution is modest and bounded: a conditional specialisation of the singular series, resolved and verified factor by factor.

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